

AGV Fleet Monitor 3D & Digital Twin



A Cyber-Physical System Integrating Edge Microcontrollers, Real-Time WebSockets, and Three.js 3D Virtualization.

PROJECT: Physical AI - ABA Fusion (Semaine 2)

FINAL DELIVERY: Jour 5 (Technical Report)

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1. Executive Summary

Modern warehouse automation relies extensively on Automated Guided Vehicles (AGVs) to coordinate logistics. Managing a physical fleet of AGVs presents challenges, particularly around system visibility, network latency, and collision prevention. This project resolves these issues by implementing a cyber-physical digital twin architecture.

By combining an embedded ESP32 microcontroller, a Python simulation routing broker, and an interactive React Three Fiber 3D client, we provide operators with real-time telemetry streaming, visual route coordinates, and manual trajectory overrides. Security is enforced via a two-layer safety mechanism combining automated hardware interrupts and cloud coordinated software locks.

2. System Architecture

The system uses a four-layered Cyber-Physical Architecture designed to ensure decoupling, reliability, and real-time visualization. Data flows from sensors up to the browser visualizer and down from the user HUD as commands.

SYSTEM ARCHITECTURE (DIAGRAMME D'ARCHITECTURE)

Embedded Edge Layer (ESP32 Microcontroller)

Reads HC-SR04 ultrasonic sensor and controls L298N motor driver.

↕ (SSL MQTT Port 8883 / Local TCP Port 1883)

Communication Bridge (HiveMQ Cloud / Mosquitto Local)

Brokers telemetry topics ("robot/distance", "robot/angle") and command queues ("robot/control"). Supports both secure TLS Cloud and offline Local Network modes.

↕ (Background Telemetry Queue Worker Threads)

Simulation & Routing Engine (Python Backend Server)

Computes Dijkstra path planning, manages Zone X Mutex lock, hosts WebSocket server, and generates dynamic English TTS audio via `en-US-JennyNeural`.

↓ (10Hz WebSocket JSON Broadcast + Base64 MP3 Speech Payloads)

3D Presentation HUD (Vite + React + R3F / Python Tkinter)

Renders animated AGVs with rotating Scanners, or displays local Tkinter Dashboards for manual override. Plays voice announcements via Web Audio API.

2.1. Hybrid MQTT Strategy (Cloud & Local Fallback)

To ensure maximum resilience and adaptability, the communication bridge has been upgraded to support a **Dual-Broker Architecture (Optional)**. Depending on the deployment environment, operators can easily toggle between two modes:

- **Cloud Mode (HiveMQ):** Uses secure SSL/TLS (Port 8883) for remote, global SCADA monitoring and digital twin synchronization. Ideal for high-level fleet management.
- **Edge/Local Mode (Mosquitto):** Uses a local standard TCP connection (Port 1883) without SSL overhead. This provides ultra low-latency control and ensures the AGVs remain fully operational even if the internet connection is completely severed. A dedicated Python Tkinter dashboard was developed specifically for direct local manual overrides.

3. Embedded Edge Layer (ESP32 Firmware)

The physical vehicle is governed by an ESP32 edge microcontroller executing a non-blocking scheduling loop. The firmware maintains strict timing for telemetry updates, LCD refreshes, ultrasonic distance scans, and MPU6050 gyroscope tracking to maximize responsiveness.

3.1. Non-Blocking Design, MPU6050 Gyroscope, and Structured JSON Telemetry

Unlike legacy looping structures that rely on `delay()`, our firmware uses `millis()` timer comparison thresholds. The telemetry system has been completely modernized to transmit data as structured JSON payloads (e.g., `{"dist":12.5, "yaw":45.2, "status":"ACTIVE"}`), greatly improving backend parsing reliability.

We integrated an **MPU6050 6-Axis Gyroscope** to track the vehicle's precise Z-axis (Yaw) rotation in real-time. This provides highly accurate heading data via MQTT, which is used to execute perfect one-shot turns without relying on blind time delays.

4. Software & Coordination Layer (Python Backend)

The Python server (running `headless_simulator.py`) manages fleet physics and navigation logic. It handles path routing and coordinates traffic intersections to prevent collisions.

4.1. Dijkstra Path Planning

The warehouse is defined as a graph containing node coordinates (Zones A, B, C, D, Charging Zone R) and lanes. When a new target is dispatched, the backend computes the shortest path using Dijkstra's algorithm and streams coordinates step-by-step.

4.2. Traffic Coordination (Zone X Mutex)

Zone X is a central intersection prone to collisions. Access is governed by a software ****Mutex Lock**** (`ZoneXTrafficController`). If AGV-01 occupies the zone, a lock is

acquired, and any secondary AGVs requesting access are placed in a `WAIT` state. Once AGV-01 exits, the lock is released, and the next vehicle proceeds.

4.3. Dynamic Pathfinder & Obstacle Avoidance (Dijkstra Re-routing)

When a vehicle (physical or simulated) encounters an obstacle, it reports the incident to the Python backend server. The server immediately closes this lane segment programmatically, triggering a real-time recalculation of the path using Dijkstra's algorithm. This closes the edge-to-server control loop, rerouting the vehicle safely around the obstacle.

5. Real-Time 3D Digital Twin Visualizer

The React frontend connects via WebSockets to the Python backend to synchronize the virtual and physical states. It features an interactive **Glassmorphism HUD** panel built with CSS backdrop-filters, custom Outfit/Space Grotesk typography, and neon glows.

Inside the Three.js 3D canvas (React Three Fiber), the visual assets have been enhanced to match the physical states:

- **Spinning Ultrasonic Scanner:** A 3D scanning cylinder is rendered on top of the AGV model and spins continuously using the animation loop to represent the HC-SR04 ultrasonic sweeps.
- **Dynamic Status LEDs:** An LED emitter changes colors based on the AGV's state: Green (moving), Orange (waiting/yielding), Red (emergency stop), and Rose (critical intersection).
- **Floor PointLight Projection:** A PointLight projects status-colored light on the floor grid under the AGV, creating real-time visual shadows.
- **Emissive Nodes:** Blue glowing spheres indicate junction coordinates.
- **First-Person View (FPV) Camera:** Operators can toggle FPV camera tracking, positioning the view slightly behind and above the AGV group, oriented dynamically to face the vehicle's driving direction.
- **Pulsing Blocked Lanes & Warning Signs:** Path segments marked as blocked render as pulsing, semi-translucent red hazard ribbons on the floor grid with 3D warning triangles that automatically face the camera.

- **English Neural TTS Alerts:** All voice announcements are generated server-side using Microsoft Edge TTS (`en-US-JennyNeural` neural voice), encoded as Base64 MP3, and streamed through the WebSocket for browser playback. This replaces the unreliable browser-level `speechSynthesis` API, delivering consistent, high-quality and natural English speech for all fleet events.
- **Danger Zone Visual Shake Feedback:** A continuous full-screen shake animation triggers when the AGV enters the precise physical boundaries of the critical intersection (Zone X). This provides immediate visual warning, and instantly ceases the moment the vehicle clears the boundary.
- **Clean Professional Dashboard:** The user interface has been streamlined, fully localized to English, and stripped of unnecessary manual controls (such as manual bypass toggles or discrete gyro commands) to emphasize autonomous operation and cleaner UX.

5.1. English TTS Technical Flow

When any fleet event occurs (E-Stop, dispatch, obstacle, Zone X entry, etc.), the Python backend calls the `announce(text)` coroutine. This function invokes `generate_english_speech(text)`, which uses the `edge_tts.Communicate` API to synthesize an MP3 file using the highly interactive `en-US-JennyNeural` Microsoft voice. The audio bytes are Base64-encoded and broadcast to all connected WebSocket clients as a special `{"type": "speech", "audio": "..."}` payload. The React frontend detects this payload type in its `ws.onmessage` handler and plays it using `new Audio("data:audio/mp3;base64," + payload.audio).play()`.

5.2. Autonomous "Square Runner" Mission

The 3D Dashboard features a dedicated **"Auto Mission"** system capable of autonomous sequence execution. We implemented a **"Square Runner"** sequence that drives the vehicle in a perfect square by looping four times through a configurable forward driving period and a rotation period. The progress of these maneuvers is dynamically visualized in the HUD via animated progress bars and countdown timers synchronized with the physical robot's movements.

Python Orchestration Logic:

```
def execute_square():
    global is_square_running
    is_square_running = True

    for side in range(1, 5):
        if not is_square_running: break
```

```
# 1. Forward movement (using BACKWARD due to inverted edge logic)
send_command("BACKWARD")
time.sleep(time_forward)

if not is_square_running: break

# 2. Right 90° Turn (MPU6050 precise rotation)
send_command("RIGHT")
time.sleep(time_turn)

# Sequence complete
send_command("STOP")
is_square_running = False
```

ENGLISH VOICE EVENTS REFERENCE

-  **E-Stop Activated:** *"Emergency stop engaged. Entire fleet has been halted."*
-  **E-Stop Cleared:** *"Emergency stop disengaged. Fleet operations resumed."*
-  **Fleet Paused:** *"Fleet operations are currently paused."*
-  **Fleet Resumed:** *"Fleet operations have resumed."*
-  **Zone Dispatch:** *"AGV 1 is now en route to zone [Zone]."*
-  **Zone X Entry:** *"Caution, vehicle has entered the critical intersection zone X."*
-  **Obstacle Detected:** *"Attention, obstacle detected ahead. Recalculating path."*
-  **Lane Re-routing:** *"Path blocked. Rerouting using Dijkstra algorithm."*
-  **Obstacles Cleared:** *"Obstacles cleared. The path is now unobstructed."*

6. Cyber-Physical Sequence & Data Flow

The diagram below traces the communication lifecycle when an operator triggers a route dispatch, highlighting the edge-level safety interrupts.

SYSTEM SEQUENCE FLOW (FLUX DE DONNÉES)

Route Dispatch

1

Operator clicks Zone A on React HUD. Web client sends `{"command": "dispatch", "target": "A"}` via WebSocket to Python backend.

Path Planning

2

Python server runs Dijkstra's algorithm to compute shortest path and starts updating motion steps.

Hardware Instruction

3

Python publishes command string (e.g. `"FORWARD"`) to the HiveMQ MQTT broker on topic `"robot/control"`.

Edge Execution & Scan

4

ESP32 receives command via Wi-Fi and triggers motor pins, while checking proximity using the HC-SR04 sensor.

Edge Safety Interrupt (Distance < 20cm)

5

If an obstacle is detected, ESP32 overrides motor speed to 0 immediately and publishes emergency telemetry to MQTT.

Digital Twin Sync

6

Python backend reads MQTT warning telemetry, stops simulated motion, and broadcasts `emergency_stop: true` to React client.

Dashboard Visual Warning

7

React displays red flashing alert overlay. Virtual AGV status LED shifts to Red, casting a red PointLight glow on the grid.

Dynamic Dijkstra Re-routing

8

Python backend registers the obstacle, programmatically closes the lane, notifies the React client (which renders the lane red with 3D warning flags), and dynamically recalculates a Dijkstra route to guide the vehicle around the blocked area.

English Neural TTS Announcement

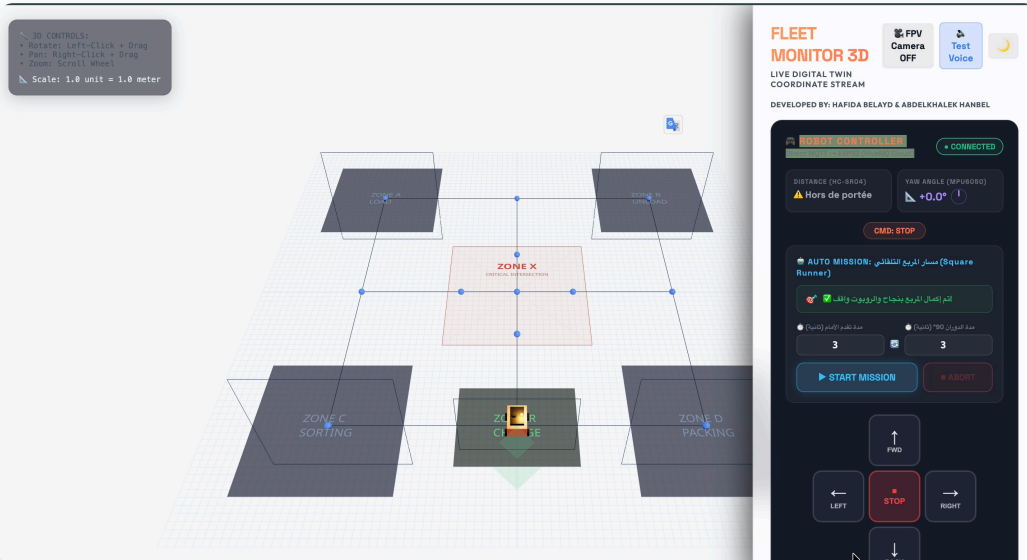
9

Python backend generates MP3 audio using `en-US-JennyNeural` Edge TTS voice, encodes as Base64, and broadcasts as `{"type": "speech", "audio": "..."} payload. Browser plays the audio automatically via Web Audio API.`

7. System Interface & Visual Analysis

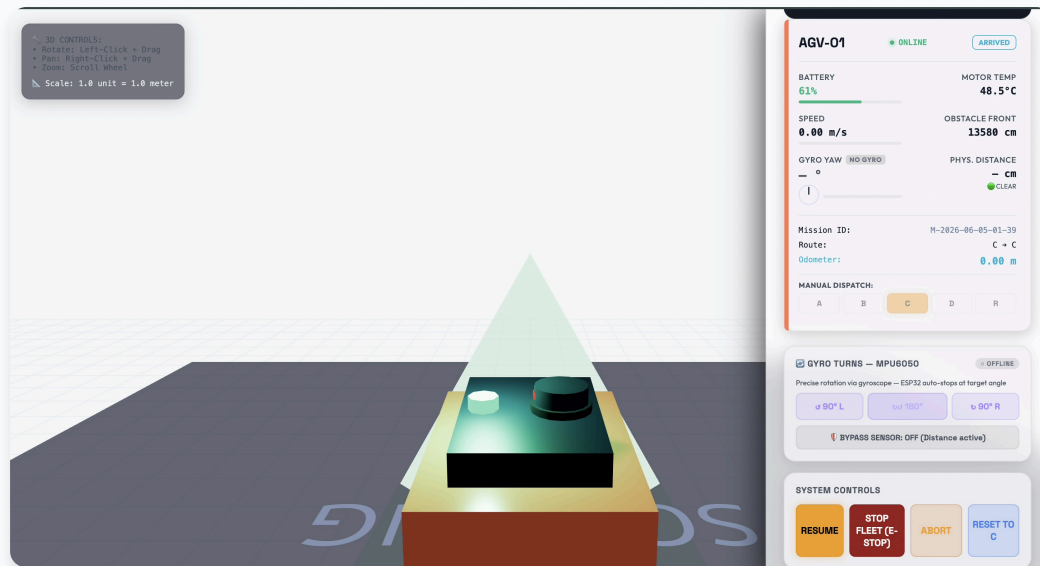
The following snapshots demonstrate the various operational states of the 3D Digital Twin and Fleet Monitor interface during real-time testing.

1. Square Runner Autonomous Mission



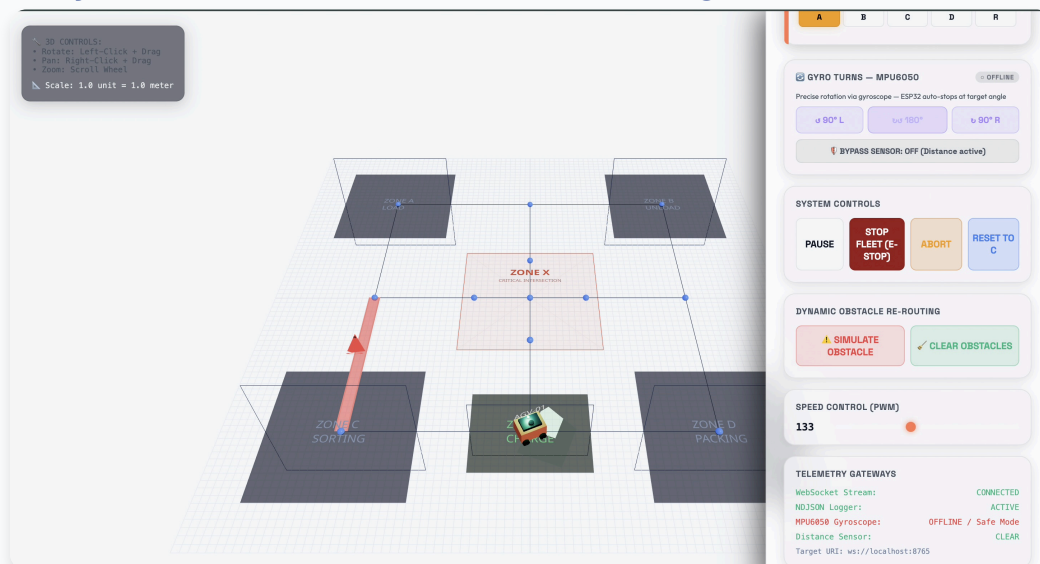
Analysis: This view highlights the "Auto Mission: Square Runner" execution pane. The interface shows a successfully completed autonomous square pattern with dynamic countdown timers synchronized for both Forward movement and 90° Gyroscope turns. The system correctly tracks time intervals mapped to real-world physics.

2. Telemetry and Manual Dispatch



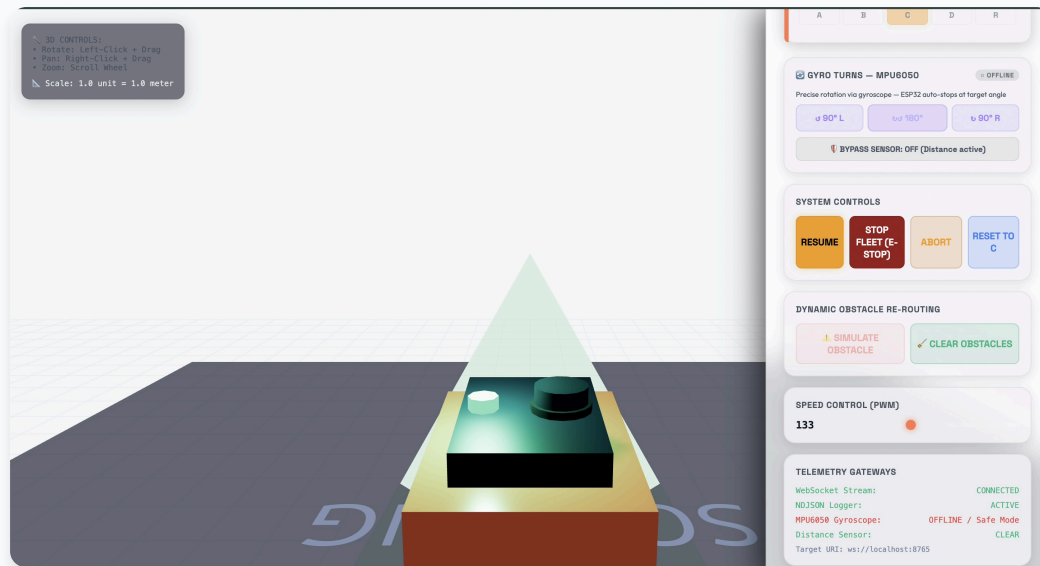
Analysis: The main Fleet Monitor layout displaying active telemetry. The sidebar tracks Live Battery (61%), Distance Sensor values, and precise Gyro Yaw angles. The 3D view shows the vehicle positioned at Zone C with realistic dynamic lighting and shadows.

3. Dynamic Obstacle Detection & Rerouting



Analysis: Shows the system's reaction to an obstacle. The route between Zone C and Zone X is dynamically marked in red (blocked), and the system highlights the active hazard using visual alerts on the 3D grid, ready to recalculate an alternate path using Dijkstra's algorithm.

4. FPV Camera System

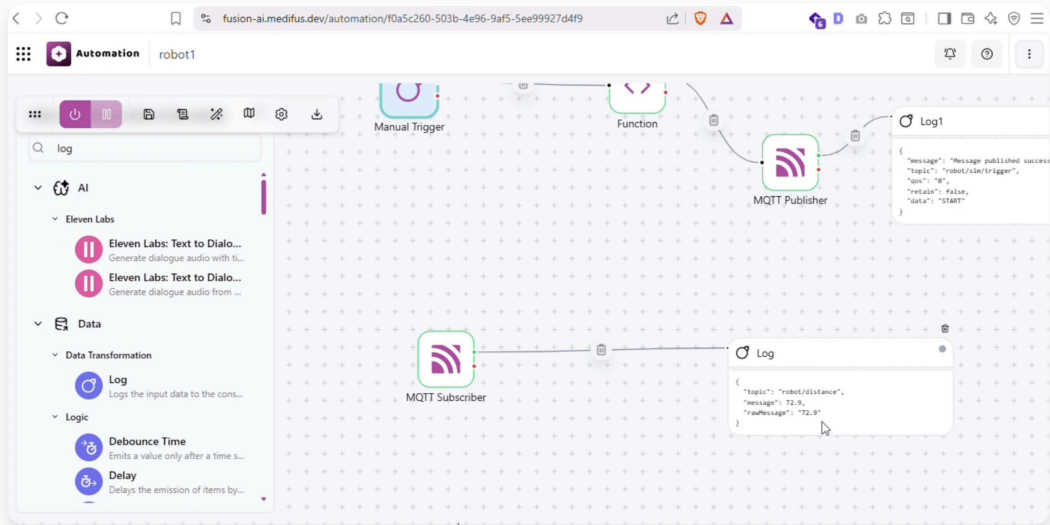


Analysis: First-Person View (FPV) Camera mode activated. The camera tightly follows the AGV from behind, showing the rotating ultrasonic scanner cone and emissive LED lights from an operator's perspective on the ground.

8. ABA Fusion Cloud Integration

To complete the Cyber-Physical Architecture, the system is fully integrated with the **ABA Fusion Cloud Automation Platform**. The node-based visual workflow (shown below) bridges the gap between manual remote operations and automated cloud logging.

Fusion MQTT Orchestration Flow



Workflow Analysis: This ABA Fusion automation pipeline serves a dual purpose:

- 1. Top Flow (Remote Dispatching):** Features a "Manual Trigger" connected to a "Function" node and an "MQTT Publisher". When triggered, it securely publishes a JSON payload with the data "START" to the robot/sim/trigger topic. This acts as a cloud-based remote control mechanism to dispatch the AGV from anywhere.
- 2. Bottom Flow (Telemetry Monitoring):** Features an "MQTT Subscriber" listening to the robot/distance topic. It captures real-time distance sensor readings (e.g., 72.9 cm) sent by the ESP32. The subsequent "Log" node records these values in the cloud, forming the foundation for historical data analysis and predictive maintenance alerts.

9. Conclusion

By integrating edge hardware safety overrides with an interactive 3D virtual twin and a professional English Neural TTS voice system, this project delivers a highly secure, traceable, and accessible AGV fleet coordination platform. The low-latency MQTT/WebSocket synchronization (upgraded to structured JSON payloads), real-time haptic-like visual warnings, and spoken English announcements powered by the en-US-JennyNeural Microsoft Edge TTS voice ensure safe and professional industrial automation.